

Option Pricing Beyond Black–Scholes

A Unified Framework: LSM, OHMC, Data Assimilation, and Coherent Inference

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1 Option Pricing Beyond Black–Scholes

Jean-Philippe Bouchaud & Marc Potters (1999, 2001)

1.1 Why These Papers Exist at All

Classical option theory teaches a remarkable idea: *you can eliminate risk completely*, and therefore derive a **unique price**.

Bouchaud and Potters begin by saying—politely, but firmly—that this miracle is fragile.

The Black–Scholes world relies on two assumptions that are rarely true together:

1. Price increments are **Gaussian at all time scales**.
2. Hedging can be done **continuously**, with no delay, no jumps, no liquidity effects.

If either assumption fails, **risk does not vanish**. And once risk does not vanish, prices are no longer unique.

This observation motivates the entire program of the 1999 paper:

What if we price options using the true, historical probability distribution, and explicitly acknowledge residual risk?

The 2001 article then sharpens the knife:

Black–Scholes is not wrong. It is just an exceptional special case that hides everything interesting about options.

1.2 Markets Are Incomplete — and That’s the Point

A market is **complete** if every contingent payoff can be replicated exactly. In such a world:

- Risk can be eliminated
- Arbitrage arguments pin down a unique price
- Historical probabilities are irrelevant

Bouchaud and Potters argue that **real markets are incomplete**, because:

- Volatility is stochastic
- Returns are fat-tailed
- Hedging is discrete
- Liquidity is finite

In an incomplete market:

- Risk remains after hedging
- Prices depend on *how* risk is measured
- Option markets exist *because* risk can be traded

“No risk, no market” is not a slogan—it is a theorem about why options exist at all.

1.3 The Core Framework: Risk Minimization

Instead of arbitrage, Bouchaud and Potters start with a **risk functional**.

They consider a trader who:

- Writes a European option
- Dynamically trades the underlying
- Ends with a random terminal P&L

Define:

- ΔW : terminal wealth change
- Risk = variance of ΔW

$$R = \langle (\Delta W)^2 \rangle$$

The goal is not to kill risk (impossible), but to **minimize it**.

This leads to:

1. An **optimal hedging strategy**
2. A **fair price**, defined by zero expected P&L
3. A **residual risk**, which does not vanish in general

This is a conceptual inversion:

Price is no longer fundamental. Risk is.

1.4 When Black–Scholes Reappears (and Why That Matters)

Under very special conditions:

- Gaussian increments
- Constant volatility
- Continuous hedging

A mathematical identity causes the residual risk to collapse to zero.

At that exact point:

- The optimal hedge becomes the delta hedge
- The price becomes Black–Scholes
- The average return disappears entirely

This explains *why* Black–Scholes works when it works—not because of arbitrage magic, but because Gaussian statistics create a cancellation miracle.

This is not a philosophical statement. It is a limit theorem.

1.5 The Average Return Is Not Irrelevant in General

One of the strangest features of Black–Scholes is that the stock’s drift does not matter.

Bouchaud and Potters show:

- This independence is **not generic**
- It fails once returns are non-Gaussian

Using a cumulant expansion of the historical distribution, they show:

- Higher moments (especially kurtosis) modify prices
- Positive drift lowers OTM call prices and raises ITM prices
- The “risk-neutral measure” becomes a **pseudo-distribution**, not a true probability

This is a profound result:

Risk-neutral pricing is an emergent approximation, not a law of nature.

1.6 Volatility Smile = Implied Kurtosis

The most empirically powerful section of the 1999 paper connects theory to data.

Key insight: The volatility smile is not mysterious. It is the market’s way of encoding **fat tails**.

They show that, to leading order:

$$\sigma_{\text{imp}}(K, T) \approx \sigma \left[1 + \frac{\kappa_T}{24} \left(\frac{(K - S_0)^2}{\sigma^2 T} - 1 \right) \right]$$

Where κ_T is the kurtosis at maturity T .

Empirically:

- Implied volatility tracks historical volatility
- Implied kurtosis tracks historical kurtosis
- Kurtosis decays anomalously slowly due to volatility clustering

The market is not irrational. It is statistically literate.

1.7 Residual Risk Is Large — and Actionable

A devastating empirical fact appears in both papers:

For a one-month at-the-money option:

Residual risk **25–30% of the option premium**

Black–Scholes says this risk is zero. Reality says it is dominant.

This matters because:

- Bid–ask spreads reflect residual risk
- Hedging errors accumulate

- Portfolio-level risk is not additive ($0 + 0 \neq 0$ anymore)

These are not academic subtleties. They are the daily problems of option desks.

1.8 Why This Changes How Option Theory Should Be Taught

The 2001 essay is pedagogical dynamite.

Bouchaud and Potters argue:

- Teaching Black–Scholes first teaches a lie
- Students learn that risk “disappears”
- Then must unlearn this in practice

Their proposal:

1. Start with generic, incomplete markets
2. Introduce hedging as risk reduction, not elimination
3. Show Black–Scholes as a degenerate limiting case

This mirrors physics education:

- Teach interacting systems first
- Introduce ideal gases later

Not elegance first—**truth first**.

1.9 Strategic Implications (Why Traders Should Care)

These papers quietly justify several modern practices:

- **Monte Carlo with historical distributions**
- **Risk-based pricing**, not arbitrage pricing
- **Statistical arbitrage** in option markets
- **Endogenous bid–ask spreads**
- **Model-based risk provisioning for OTC products**

They also explain why:

- Markets revert to payoff at maturity
- Mispricings cannot persist forever
- Options encode forward-looking beliefs about higher moments

This is option pricing as *Bayesian inference under uncertainty*—even if the authors never use that language explicitly.

1.10 The Big Picture

Bouchaud and Potters do not “improve” Black–Scholes. They **demote it**.

Black–Scholes becomes:

- A beautiful
- Rare
- Zero-risk
- Gaussian
- Continuous-time limit

The real world lives outside that limit.

And once you accept that, option theory stops being about elegance and starts being about **judgment under uncertainty**—which is exactly what markets are for.

Closing Thought: Black–Scholes tells you what happens in a world without storms. Bouchaud and Potters teach you how to sail. And markets, inconveniently, have weather.

2 Monte Carlo Methods: LSM and OHMC

2.1 The Big Idea

Option pricing is a backwards-in-time problem: at maturity you **know** the payoff, and earlier you **don't**. Both LSM and OHMC turn that backwards recursion into a sequence of **linear regressions** on simulated (or historical) paths.

But they have very different philosophies:

LSM (Longstaff–Schwartz style): Primarily a way to decide **when to exercise** (American options). The regression estimates the *continuation value*; the final price is still basically an average of discounted cashflows along paths. For a **European** option it collapses to standard Monte Carlo.

OHMC / HMC (Potters–Bouchaud–Šestović): Builds the hedge *into* the regression by minimizing a **local hedging error** (a risk functional). This simultaneously estimates:

1. the option price function,
2. the hedge function, and
3. the residual risk,

and yields a **huge variance reduction** because hedging kills variance.

Think of OHMC as: *“Monte Carlo, but with a control variate that is the hedged P&L — and the control variate is chosen optimally.”*

2.2 Shared Setup: Discrete-Time Paths and Backward Induction

The data setup for both methods:

- Many paths $x_{l,k}$: path index $l = 0, \dots, m$, time index $k = 0, \dots, n$
- Time step $\Delta t = T/n$
- At maturity $k = n$: $C(x_{l,n})$ is just the payoff (for a call: $\max(x_{l,n} - K, 0)$)

The core numerical trick is always the same:

Represent the unknown function $C_k(x)$ using basis functions, then fit coefficients by regression on the cloud of simulated points.

2.3 Least Squares Monte Carlo (LSM)

2.3.1 What LSM Is Minimizing

The risk-minimization lens motivates LSM:

Global form: minimize squared pricing error between discounted payoff and today's price:

$$R = (C(x_n)e^{-rT} - C(x_0))^2$$

Local stepwise version:

$$R_k = (C(x_{k+1})e^{-r\Delta t} - C(x_k))^2$$

Approximate $C(x_k)$ by a linear combination of basis functions:

$$C(x_k) = \sum_{i=0}^M b_i^{(k)} f_i(x_k)$$

This turns each backward step into a linear least squares fit.

2.3.2 The Regression Form

The regression in matrix form uses the normal equations estimator:

$$\hat{B}_k = (X_k^\top X_k)^{-1} X_k^\top Y_k$$

where:

- Y_k is “next-step value discounted back one step”
- X_k is the design matrix of basis functions evaluated at $x_{t,k}$

2.3.3 LSM Algorithm

1. Compute payoffs at maturity to get $\hat{C}_n$
2. For $k = n - 1, \dots, 0$:
 - Discount values to form Y_k
 - Build X_k from current prices
 - Estimate coefficients $\hat{B}_k$ via least squares
 - Predict $\hat{C}_k$
3. Arrive at $\hat{C}_0$, then average across paths for the initial price estimate

2.3.4 Where LSM Shines (and Where It Doesn't)

The Longstaff–Schwartz interpretation:

- The regression is primarily used to identify **early exercise**: compare intrinsic value vs continuation value
- The *final* price is the average discounted cashflow across paths
- For **European** options, “LSM is identical to standard Monte Carlo”

That sentence is a dagger, and it's true: without early exercise, LSM doesn't reduce variance much by itself.

2.4 Optimal Hedged Monte Carlo (OHMC / HMC)

Now we switch to the Potters–Bouchaud–Šestović move: **put the hedge inside the objective**.

2.4.1 The Key Object: The Local Risk Functional

The local risk (variance of hedged wealth change) is:

$$R_k = \left\langle (C_{k+1}(x_{k+1}) - C_k(x_k) + \phi_k(x_k)(x_k - x_{k+1}))^2 \right\rangle_o$$

Crucially, the brackets are over the **objective (historical) measure**, not risk-neutral.

This one equation is the whole philosophy:

- You’re not pretending you know the risk-neutral measure
- You’re choosing a hedge to minimize local hedging error
- The pricing emerges from a “fair game” condition after risk minimization

2.4.2 Basis Expansions for Both Price and Hedge

Expand **both** $C_k(x)$ and $\phi_k(x)$ in basis functions:

$$C_k(x) = \sum_a \gamma_a^{(k)} C_a(x), \quad \phi_k(x) = \sum_a \varphi_a^{(k)} F_a(x)$$

The backward step becomes: choose coefficients to minimize a quadratic loss across all Monte Carlo paths (a linear least squares problem).

2.4.3 Why Variance Drops So Hard

Minimizing financial risk by optimal hedging *also* minimizes Monte Carlo variance — same mechanism, different language.

The practical punchline: standard deviation often **5–10× smaller**, meaning you may need up to **~100× fewer paths** for the same precision.

This isn’t magic. It’s control variates, but with the best possible control variate available: the hedged P&L.

2.4.4 Objective vs Risk-Neutral: The “Quiet Conversion”

A subtle but profound claim:

“The explicit accounting of the hedging cost naturally converts the objective probability into the ‘risk-neutral’ one.”

Operationally: you can simulate under the historical distribution (or even *use historical time series directly*) and still end up with a consistent pricing scheme because the hedge adjustment removes drift dependence.

This is why OHMC is attractive for real markets where the “true” process is messy: fat tails, volatility clustering, leverage effect, etc.

2.5 OHMC vs LSM: The Clean Comparison

Feature	LSM	OHMC
Probability measure	Risk-neutral (fixed)	Objective / historical
Hedge in regression	No	Yes
Variance reduction	Minimal for Europeans	5–10× improvement
Output	Price only	Price + hedge + residual risk

Feature	LSM	OHMC
American options	Primary use case	Also supported

Key insight: LSM carries forward the pathwise discounted payout/exercise value; regression is mostly an exercise classifier. OHMC regression is a **hedged** regression (price + hedge coefficients).

2.6 Algorithm Sketches

2.6.1 LSM (European or American)

1. Initialize: $V_{l,n} = \text{payoff}(x_{l,n})$
2. For $k = n - 1$ down to 0:
 - Regress $e^{-r\Delta t}V_{l,k+1}$ on basis $f_i(x_{l,k})$
 - Get predicted continuation $\hat{C}_{l,k}$
 - If American: set $V_{l,k} = \max(\text{intrinsic}(x_{l,k}), \hat{C}_{l,k})$; else $V_{l,k} = \hat{C}_{l,k}$
3. Price $\approx \frac{1}{m+1} \sum_l V_{l,0}$

2.6.2 OHMC / HMC (European or American)

1. Initialize: $C_{l,n} = \text{payoff}(x_{l,n})$
2. For $k = n - 1$ down to 0:
 - Choose basis for price $C_a(x)$ and hedge $F_a(x)$
 - Fit coefficients $(\gamma^{(k)}, \varphi^{(k)})$ minimizing the hedged squared error across paths
 - Compute fitted $C_k(x_{l,k})$ and $\phi_k(x_{l,k})$
 - If American: overwrite pathwise values where exercise dominates
3. Output: price function $C_0(\cdot)$, hedge function $\phi_0(\cdot)$, and residual risk

2.7 Implementation Gotchas

Both methods live or die on:

- **Basis functions** (too weak $\rightarrow$ bias; too flexible $\rightarrow$ instability)
- **Data coverage** (you need enough paths across the relevant state region)
- For OHMC specifically: **hedge basis** choices (hedge misspecification leaves variance on the table)

You can constrain the hedge to be the derivative of the price basis (a delta-like restriction) to cut computation cost, but it's only exact in Gaussian settings.

3 Coherent Prevision: Lad's Framework

3.1 Chapter 2 — Quantities, Prevision, and Coherency

3.1.1 Why Chapter 2 Is the True Foundation

Most probability texts start with sample spaces, sigma-algebras, and axioms. Lad starts with **behavior**.

Chapter 2 answers one question:

If probabilities are *assertions of uncertain knowledge*, what rules must they obey so that a person cannot be made a guaranteed loser?

This is not about metaphysics. It is about **self-consistency under action**.

3.1.2 Quantities: What We Are Allowed to Talk About

A **quantity** is:

- Something whose value is *unknown to you now*
- But will be *revealed* by a well-defined procedure

Examples: tomorrow's temperature, the payoff of an option at maturity, whether an event occurs (coded as 0/1).

Crucial insight:

Probability is not defined over “events.” Probability is defined over **quantities**.

Events are just special quantities that take values $\{0, 1\}$.

3.1.3 Prevision: The Primitive Notion

Instead of “expectation,” Lad uses **prevision**.

Definition (informal): Your **prevision** for a quantity X , written $P(X)$, is the price at which you are willing to buy or sell X in a fair contract.

This is **operational**:

- It involves money
- It involves commitment
- It can be tested

No metaphysics. No “true” expected values. Just: *what are you willing to stand behind?*

3.1.4 Why Prevision Is Stronger Than Expectation

Expectation in classical probability:

- Is defined relative to a probability measure

- Presumes a distribution exists

Prevision:

- Comes **before** distributions
- Applies to any quantity, whether or not it is repeatable
- Does not require a sample space

This reverses the usual order:

Distributions do not define expectations. Previsions define distributions.

3.1.5 Coherency: The Core Constraint

Your set of previsions is **coherent** if there is no combination of bets that guarantees you a loss, no matter what happens.

This is the **Dutch Book principle**, but Lad frames it geometrically and algebraically rather than morally.

Key insight:

Incoherence is not “being wrong.” Incoherence is **logical contradiction under action**.

If your beliefs are incoherent, someone can exploit you **with certainty**.

3.1.6 Linear Structure of Prevision

Coherency forces prevision to obey **linearity**.

For quantities X, Y and real a, b :

$$P(aX + bY) = aP(X) + bP(Y)$$

This is not assumed. It is **derived** from the requirement of no sure loss.

Expectation linearity is not a modeling choice. It is a *logical necessity of coherent belief*.

3.1.7 Events as Quantities: Probability Recovered

Let E be an event. Define the indicator quantity:

$$I_E = \begin{cases} 1 & \text{if } E \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

Then Lad defines:

$$P(E) := P(I_E)$$

Probability is now a special case of prevision, not a primitive.

Kolmogorov’s axioms become **theorems**, not assumptions.

3.1.8 The Fundamental Theorem of Prevision (FTP)

FTP (Part I): A set of previsions is coherent **if and only if** it can be extended to a linear prevision over all bounded quantities without creating a sure loss.

Interpretation:

- Coherence is not fragile
- Partial beliefs can be extended consistently
- Learning is logically possible

This theorem is the bridge from belief to inference.

3.2 Chapter 3 — Coherent Statistical Inference

3.2.1 Conditional Prevision: The Primitive of Inference

Lad does **not** start with conditional probability. He starts with **conditional prevision**.

Definition (operational): Your **conditional prevision** of a quantity X given event H , written $P(X | H)$, is the price you regard as fair for the contract “receive X *if* H occurs; otherwise the contract is canceled.”

Key points:

- No division by $P(H)$
 - No assumption that $P(H) > 0$
 - Conditioning is a *contractual restriction*, not a ratio
-

3.2.2 Bayes’ Theorem as a Coherence Identity

Lad derives Bayes’ theorem **without assuming probability measures**.

For events A and B , coherence implies:

$$P(A | B) \cdot P(B) = P(B | A) \cdot P(A)$$

This is not an axiom. It is an **accounting identity** forced by coherent prevision.

Bayes’ theorem is not “how rational people update.” It is **how incoherence is avoided**.

3.2.3 Total Prevision: Law of Iterated Expectation

Lad derives the subjective version of the law of total expectation:

$$P(X) = P(P(X | H))$$

Interpretation: today’s prevision equals today’s prevision of tomorrow’s prevision.

This is about **time consistency of belief**. Violating this identity means you are guaranteed to reverse yourself in a way that can be exploited.

This result is the backbone of dynamic programming, Bellman equations, and martingale pricing logic—all without stochastic processes.

3.2.4 Exchangeability: The Substitute for Independence

Definition (informal): A sequence of events is **exchangeable** if your previsions are unchanged by permuting their order.

Exchangeability:

- Does **not** assume independence
- Does **not** assume identical distributions
- Encodes a *judgment of symmetry in ignorance*

Independence is a mathematical convenience. Exchangeability is a **belief**.

3.2.5 De Finetti’s Representation Theorem (Conceptual Role)

If you regard events as infinitely exchangeable, then your beliefs are equivalent to a mixture over i.i.d. models.

This is not saying “there exists a true underlying probability.”

It is saying: “your beliefs behave *as if* such a probability were drawn and then held fixed.”

Probability distributions become **summaries of belief**, not generators of reality.

3.3 Chapter 4 — Related Forms for Asserting Uncertainty

3.3.1 Prevision Is Primary — Everything Else Is Derived

Means, variances, quantiles, correlations, and moments are **not primitives**. They are *derived summaries* of coherent previsions.

This reverses standard statistics, where moments are assumed, distributions are fitted, and coherence is checked (if at all) later.

3.3.2 Variance as a Prevision of Loss

Lad defines variance as:

$$\text{Var}(X) = P((X - P(X))^2)$$

This reframes variance as:

- A **prevision of quadratic loss**
- Not a property of nature

Variance measures how painful it is to be wrong *under a specific scoring rule*.

3.3.3 Coherency Restrictions on Variance and Covariance

Not every collection of means and variances is coherent.

Examples:

- Variance must be non-negative
- Covariance matrices must be positive semi-definite
- Correlations must lie in feasible geometric regions

These are **logical constraints**, not statistical assumptions.

A person asserting incompatible variances is incoherent—they can be Dutch-booked.

3.4 Chapter 6 — Proper Scoring Rules

3.4.1 What a Scoring Rule Is

A **scoring rule** assigns a numerical score after the outcome is observed, based on:

- The forecast (probability or prevision)
- The realized outcome

Scores represent: reward, penalty, loss, reputation, capital.

Scoring rules turn beliefs into **stakes**.

3.4.2 Proper Scoring Rules

A scoring rule is **proper** if your expected score is maximized (or loss minimized) by stating your true prevision.

If a scoring rule is improper:

- It incentivizes lying
 - It distorts belief reporting
 - It destroys the operational meaning of probability
-

3.4.3 Quadratic Scoring Rule

For a quantity X :

$$S(P, X) = -(X - P)^2$$

Key properties:

- Expected score maximized at true prevision
 - Variance emerges as expected loss
 - Links directly to OHMC
-

3.4.4 Logarithmic Scoring Rule

For positive quantities or distributions:

$$S(p, x) = \log p(x)$$

This leads to entropy, KL divergence, and likelihood-based inference.

But likelihood is not sacred—it is just one proper scoring rule.

Maximum likelihood = expected log score maximization. That’s it. No metaphysics.

3.4.5 Calibration vs Refinement

Scoring performance decomposes into:

- **Calibration:** are stated probabilities consistent with frequencies?
- **Refinement:** are predictions sharp or vague?

Crucially:

Coherent previsions are *automatically calibrated*.

Poor calibration implies incoherence or dishonesty.

3.4.6 One-Sentence Summary

A probability statement means nothing unless there exists a rule that makes honesty optimal.

4 The Grand Synthesis

4.1 Lad Bouchaud–Potters OHMC ABC

One framework: coherent beliefs + operational bets + simulation-as-inference

4.1.1 The Shared Primitive: Commitments You Can Be Dutch-Booked On

Lad’s move (operational subjectivism): Probability is a **commitment** you assert about uncertain quantities, and it must obey **coherence** (no sure-loss). Uncertainty statements only matter if they connect to decisions and stakes.

Bouchaud–Potters move (risk minimization): In most realistic models, hedging cannot eliminate risk; strict arbitrage pricing fails; residual risk matters and is a reason option markets exist. Replace “unique price from arbitrage” with: **choose price + hedge by minimizing a risk functional**.

This is Lad’s coherence principle, translated into trading language.

4.1.2 The Bridge Concept: Quadratic Scoring = Variance Risk = Least Squares

This is the deepest connection:

- Lad’s quadratic score is literally squared error
- BP’s OHMC chooses coefficients by minimizing a **quadratic** hedging loss

So OHMC is not just “a Monte Carlo trick.” It is:

Proper scoring of hedged predictions, where the “prediction” is the continuation value and the “penalty” is the squared hedging error.

4.1.3 OHMC as “Operational Probability” in Action

In Lad’s framework:

Lad Concept	OHMC Analogue
Prevision	Option value function $C_k(x)$
Decision/act	Hedge $\phi_k(x)$
Proper score / loss	Squared hedging error (local risk)
Coherence	Fair-game pricing + no systematic exploitation
Residual uncertainty	Residual risk curve

That’s exactly Lad’s “probability is operational” doctrine, but now the operational test is: *did your stated value + hedge avoid a systematic loss and minimize dispersion?*

4.1.4 ABC as the Inference-Side Twin of OHMC

ABC in one sentence: ABC is a way to do Bayesian inference when the likelihood $p(y \mid \theta)$ is intractable but **simulation is easy**.

Instead of computing the likelihood, you:

1. Sample θ from the prior
2. Simulate synthetic data $x \sim p(\cdot \mid \theta)$
3. Compare summaries $s(x)$ to $s(y)$
4. Accept θ when the discrepancy is small

Why ABC is “Lad-compatible”: Conditioning is replaced by *operational similarity*: “treat simulated worlds as compatible with reality if they match the observed summaries closely enough.”

Why ABC is “Bouchaud–Potters compatible”: Both are obsessed with *not lying to yourself* about model mismatch.

4.1.5 The Unifying Triangle: Belief Action Score

Lad’s triangle:

1. **Belief:** coherent previsions
2. **Action:** decisions/bets based on those previsions
3. **Score:** proper scoring rules evaluate quality and incentivize honesty

OHMC instantiates the triangle in markets:

1. Belief: continuation value function $C_k(x)$
2. Action: hedge $\phi_k(x)$
3. Score: hedged P&L variance (quadratic loss)

ABC instantiates the triangle in inference:

1. Belief: prior over parameters/models
 2. Action: simulate candidate worlds and filter by discrepancy
 3. Score: the discrepancy metric acts like a scoring rule
-

4.1.6 A “Single Engine” View: ABC Feeds OHMC

Step A — Use ABC to infer the objective dynamics:

You have some complex market mechanism (fat tails, volatility clustering, jumps, regime shifts). Likelihood is nasty.

- Use ABC to infer parameters θ of your simulator
- The output is an approximate posterior $p_\varepsilon(\theta \mid y)$

Step B — Price & hedge with OHMC under objective probabilities:

For each sampled θ :

- Simulate price paths under the objective dynamics $p(\cdot \mid \theta)$

- Run OHMC backward regression to obtain derivative value, hedge function, and residual risk
- Then propagate parameter uncertainty to get a *Bayesian* distribution over prices and hedging risk.
-

4.2 OHMC as a Proper Scoring Rule

4.2.1 The OHMC Score

At time step k , OHMC predicts the discounted next-step value:

$$Y_k := D_k V_{k+1}(Z_{k+1})$$

With hedging action $\phi_k(z)$, the **realized hedged target** is:

$$\tilde{Y}_k := Y_k - \phi_k(z)^\top \Delta H_{k+1}$$

The quadratic scoring rule:

$$S_k(C_k(z), \phi_k(z); Y_k, \Delta H_{k+1}) := -\left(Y_k - C_k(z) - \phi_k(z)^\top \Delta H_{k+1}\right)^2$$

Why this is proper: Because quadratic loss is strictly proper for the conditional mean:

$$C_k^*(z) = \mathbb{E}[\tilde{Y}_k \mid Z_k = z]$$

is the unique minimizer of expected squared error.

4.2.2 The Regression Step as Empirical Risk Minimization

Given simulated paths $i = 1, \dots, n$, the OHMC regression is:

$$(\hat{C}_k, \hat{\phi}_k) = \arg \min_{C, \phi} \sum_i \left(y_k^{(i)} - C(z_k^{(i)}) - \phi(z_k^{(i)})^\top \Delta h_{k+1}^{(i)} \right)^2$$

which is exactly minimizing the **empirical expected loss** associated with the proper score.

OHMC is literally: minimize empirical negative proper score for the hedged forecast.

4.3 Data Assimilation Connection

4.3.1 The Shared Skeleton

In discrete time DA, the model is:

State evolution: $x_{k+1} = \mathcal{M}(x_k) + \eta_k$

Observation model: $y_k = \mathcal{H}(x_k) + \xi_k$

The DA task is to infer $\pi(x_k \mid y_{1:k})$ (filtering) or $\pi(x_{0:k} \mid y_{1:k})$ (smoothing).

4.3.2 OHMC Reinterpretation

DA Object	OHMC Analogue
State x_k	Economic state Z_k
Model $\mathcal{M}$	Market simulator
Observation y_k	Realized discounted future value
Observation error	Unhedged P&L noise
Control	Hedge $\phi_k(Z_k)$

Key insight:

OHMC treats future cashflows as noisy observations of continuation value

4.3.3 Variational DA = OHMC with Quadratic Scoring

The 4D-Var cost function:

$$J(x_{0:T}) = \frac{1}{2} |x_0 - x_b|_{B^{-1}}^2 + \frac{1}{2} \sum_{k=1}^T |y_k - \mathcal{H}(x_k)|_{R^{-1}}^2$$

The OHMC objective:

$$\sum_k \left(Y_k - C_k(Z_k) - \phi_k(Z_k)^\top \Delta H_{k+1} \right)^2$$

They are the **same mathematical object**: quadratic penalty + model-based propagation + regression as variational minimization.

OHMC is a sequential variational data-assimilation scheme where the control variable is a hedge.

4.4 Shannon–Kelly Connection

4.4.1 Log Score Is Shannon Surprise

If you forecast a density $p(\cdot)$ and outcome y happens, the **log score** is:

$$S(p, y) = \log p(y)$$

Expected surprisal is **entropy**: $H(Y) = \mathbb{E}[-\log p(Y)]$

KL divergence is expected log-score improvement:

$$\mathbb{E}_p[\log p(Y) - \log q(Y)] = \text{KL}(p\|q) \geq 0$$

Better probabilistic models literally earn you KL in log-score currency.

4.4.2 Kelly Betting: Information → Wealth

Kelly says choose fraction f to maximize **expected log wealth**:

$$f^* \in \arg \max_f \mathbb{E}[\log(1 + fR_{t+1})]$$

Why log?

- It makes the objective **time-additive**
- It maximizes the **almost sure long-run growth rate**

4.4.3 The Shannon–Kelly Identity

Your optimal expected log growth equals mutual information.

The value of side information Y for log-optimal betting is:

$$\Delta g = I(X;Y)$$

Bits become bankroll growth under log utility.

4.4.4 Two Levels of Proper Scoring

Level	Score	Method
A: Mean-based	Quadratic	OHMC classic (variance control)
B: Distribution-based	Log	Kelly/Shannon (growth optimization)

4.5 The Final Synthesis

4.5.1 What They All Share

All these methods solve:

$$\min_{\text{policy}} \mathbb{E}[\text{future loss given uncertain dynamics}]$$

Method	State	Data	Optimizes
Data Assimilation	Physical state	Sensors	Estimation error
ABC	Parameters	Summaries	Simulation mismatch
LSM	Option value	Payoff	Price
OHMC	Value + hedge	P&L	Risk
Kelly	Wealth	Returns	Growth rate

Same machinery, different nouns.

4.5.2 The Unified Loop

simulate → infer state → value decision → hedge → observe → update belief → repeat

This loop is:

- Kalman filter
- Particle filter
- ABC
- LSM
- OHMC
- Reinforcement learning
- Trading

Same engine. Different skin.

4.5.3 One Sentence to Hold Onto

They are not the same algorithm, but they are all solutions to the same kind of problem: learning and acting optimally under uncertainty.

4.5.4 The Thesis Sentence

Optimal Hedged Monte Carlo is a sequential variational data-assimilation algorithm in which future cashflows act as noisy observations, hedging actions act as controls, and proper scoring rules replace likelihoods as the criterion for coherent belief-action updating.

This sentence is defensible simultaneously in:

- Lad’s decision-theoretic framework
- Bouchaud–Potters risk minimization
- Law–Stuart–Zygalakis DA theory
- Evensen’s ensemble formulation
- ABC / likelihood-free inference

4.6 Summary Table

Framework	Contribution
Lad	Epistemology: probability = coherent prevision
Bouchaud–Potters	Market reality: residual risk is central
OHMC	Operational bridge: price + hedge + residual risk
ABC	Inference twin: likelihood-free learning
Shannon–Kelly	Information–wealth connection
Data Assimilation	State estimation under uncertainty

Put together: ABC learns the simulator; OHMC prices/hedges inside it; Lad supplies the coherence and scoring logic that makes the whole thing intellectually honest; Shannon–Kelly connects information to growth.